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# Financial Literacy and Financial Inclusion among University Students in Jordan

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ABSTRACT

By providing empirical support for the relationship between financial literacy and financial inclusion among Jordanian university students, this study contributes considerably to the growing body of literature on financial education and inclusion. Policymakers, educators, and other stakeholders working to improve financial education programs and advance financial inclusion initiatives should consider the findings. This study highlights the significance of cultivating financial knowledge and skills among the youth population in Jordan and abroad because financial literacy continues to be a critical driver of financial well-being and economic stability. The analysis is supported by a thorough survey instrument that was carefully validated through expert consultation and a pilot study and was modified from the OECD/INFE. The resulting survey assesses FL and FI and demographic factors, including gender, major, age, and year of study. With a coefficient of 0.76, Cronbach's alpha test validates the instrument's dependability and internal consistency. In addition, an investigation of gender-based differences in FL scores, although observed, did not reach statistical significance, indicating the influence of other contextual factors. Using regression analysis and Spearman's rho correlation, this study discerns a strong and statistically significant positive relationship between FL and FI among university students in Jordan (89.4%, $P<0.01$), corroborating the central hypothesis. The study's econometric model, supported by tests for goodness-of-fit, exhibits statistical significance, underscoring its potency in clarifying the intricate dynamics of FL and FI in this situation. The study emphasizes the critical part financial literacy plays in promoting financial inclusion and the demand for customized strategies that consider the variety of students' academic backgrounds. The findings substantially contribute to the global conversation on financial inclusion and education, highlighting their critical role in fostering economic stability and progress.

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## 1. Introduction

There is a growing appreciation of the importance of Financial Literacy (FL) because of the rapid change in financial services, including the ease of access and complexity (Hastings & Mitchell, 2020). Moreover, the global financial crisis in 2008 revealed the consequences of the lack of FL among most of those dealing with financial institutions (Özdemir, Temizel, Sönmez, & Fikret, 2015), which prompted decision-makers worldwide to develop national strategies to promote FL. Therefore, in the year 2012, the leaders of the G20 countries endorsed the main principles of the National Strategies for Financial Education developed by the Organization for Economic Cooperation and Development International Network on Financial Education (OECD/INFE) (Morgan & Trinh, 2019). According to Bongomin, Ntayi, and Malinga (2020), the need for people to be financially literate is increasing due to rising consumer debt, declining savings, and increasing bankruptcy rates, especially in developing countries. FL includes an individual's knowledge of basic financial concepts, the ability to manage personal money and financial planning, and considering changing economic events (Arora, 2020; M. N. Khan, Ferrer, Lee, & Rothwell, 2022). Thus, FL provides individuals with the knowledge and skills to make rational financial decisions and evaluate financial services and products that affect personal financial welfare (Kasozi & Makina, 2021; Okello Candiya Bongomin, Munene, & Yourougou, 2020). Moreover, FL affects the country's economy and financial system stability by increasing savings and investing rates and encouraging financial institutions to innovate financial services that meet the needs of financially literate customers (Ergün, 2017).

Douissa (2020) and Oseifuah et al. (2018) stressed that there is a growing awareness of the importance of FL among young people due to the complexity of financial products and the possibility of increasing financial risks resulting from increased life expectancy, decreasing social welfare, and uncertainty in career and economic prospects. Indeed, despite global efforts in this regard, existing data show that FL is still low among youth and university students around the World (Ergün, 2017; Garg & Singh, 2018).

Financial Inclusion (FI) proxied by access to financial services remains one of the policymakers' priorities worldwide because of its essential role in achieving sustainable development goals, enhancing financial stability, and reducing poverty (Grohmann, Klühs, & Menkhoff, 2018; F. Khan, Siddiqui, & Imtiaz, 2022; Ozili, 2018). Although the efforts made in this context, the recent Findex survey indicated that 1.4 billion adults are financially excluded, most of whom live in developing countries (Demirgüç-Kunt, Klapper, Singer, & Ansar, 2022). Previous studies show that effective FI strategies consider financial services' supply and demand factors simultaneously. Demand factors are related to the users of financial services and reflect the extent to which users have the desire and ability to use financial services that satisfy their needs, while supply factors are related to the financial service providers and reflect the suitability of these services to users' needs (Abdeldayem, 2016; Atkinson & Messy, 2013). Grohmann et al. (2018) contend that the availability of financial services that

meet users' needs must be accompanied by financially literate users who can make the right financial decisions, which leads to enhancing the financial system's effectiveness and increasing the level of FI. Moreover, poor awareness of financial services and the lack of knowledge of how such services work create barriers to accessing formal financial services, negatively affecting FI (Atkinson & Messy, 2013).

Arab countries, including Jordan, have paid attention to FI as one of the determinants of financial stability, well-being, and equality. However, a low rate of FI characterizes the Arab region compared with other regions worldwide. According to Demirgüç-Kunt et al. (2022), the outreach of formal financial services in the Arab region is only 29%. Hence, the Arab Monetary Fund established the Financial Inclusion Initiative for the Arab Region in cooperation with the German Development Agency and the global alliance for FI, with the participation of the World Bank. Such initiative aims to support FI policies to expand the scope for citizens and institutions to benefit from official financial services that suit their needs. In Jordan, the Central Bank of Jordan (CBJ) launched the National Financial Inclusion Strategy (NFIS) in 2017 to foster FI at the national level. NFIS has identified four FI enablers: financial technology, data and research, regulations, and financial capability, identified as an individual's capacity to manage financial resources and make sound financial decisions, and consciously benefit from financial services. Accordingly, the CBJ has organized many initiatives to increase the FI level, targeting school and university students, women, and entrepreneurs. However, despite the efforts made in this regard, the proportion of FI in Jordan reached 47.1% in 2021, which is low compared to the World average of 76.2%, and other countries in the Arab region, such as Saudi Arabia and the United Arab Emirates (Demirgüç-Kunt et al., 2022). On the other hand, the FL level in Jordan is low. According to the Global FinLit survey developed by Standard & Poor’s in 2014, only 43% of adults in Jordan are financially literate (Klapper, Lusardi, & Van Oudheusden, 2015).

The impact of FL has been investigated in different contexts. Previous studies demonstrate the effect of FL on retirement planning (Hastings & Mitchell, 2020), saving behavior (Morgan & Long, 2020), investment decision-making (Alaaraj & Bakri, 2020), financial behavior (Allgood & Walstad, 2016), household wealth accumulation (Behrman, Mitchell, Soo, & Bravo, 2012), and stock market participating (Bucher-Koenen, Alessie, Lusardi, & Van Rooij, 2021). However, little is known about the impact of FL on FI, and there is a need for further studies in this area, especially in developing countries like Jordan (Lyons & Kass-Hanna, 2021; Mouna & Anis, 2017). Consequently, this study aims to assess the level of FL and FI among university students in Jordan and examine the relationship between FL and FI. Based on the author's knowledge, this is the first study focusing on this issue in Jordan. It can provide valuable information about FL and FI levels among university students in Jordan, which helps in comprehending the future demand for financial services that significantly contribute to economic growth and financial stability.

Moreover, it provides data for decision-makers, which is the first step to formulating FL policy. In addition, the study's outcome will increase understanding of the relationship between FL and FI. Consequently, it will provide insight for policymakers to build effective strategies to promote FI, considering the role of FL.

To achieve the study's goal of understanding the relationship between financial literacy and financial inclusion among university students in Jordan, we employed a robust methodology. We modified the OECD/INFE survey instrument for the Jordanian context, guaranteeing its validity through expert consultation and a pilot study. We investigated the relationship between financial literacy and financial inclusion using regression analysis and Spearman's rho correlation and discovered a significant positive effect. We also examined the disparities in financial literacy between men and women and evaluated the impact of academic majors on financial literacy. Thanks to this thorough methodology, we learned important new information on the variables affecting Jordanian university students' financial well-being. The paper proceeds as follows: Section 2 summarizes the previous related literature. Section 3 describes the methodology used and data collection procedures. Section 4 analyzes and discusses the results, and Section 5 reports the conclusions.

## 2. Literature Review

*2.1 Financial Literacy and Financial Inclusion: Conceptual Framework*

Goyal and Kumar (2021), who reviewed 502 published papers on FL from 2000, mentioned that no standard definition of FL coerces researchers to use terms such as financial capability, financial knowledge, financial behavior, and financial education synonymously. Huston (2010, p. 306) defines FL as "measuring how well an individual can understand and use personal finance-related information." Lusardi and Mitchell (2014, p. 7) define FL as the "ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt, and pensions." Recently, The Organization for Economic Cooperation and Development (OECD) defined FL as a "combination of awareness, knowledge, skills, attitudes, and behavior necessary to make sound financial decisions and ultimately achieve individual financial well-being" (OECD, 2020, p. 27). The OECD definition is the most used in the literature because it comprehensively covers the dimensions of FL: financial behavior, financial knowledge, and financial attitude. Morgan and Long (2020) argued that financial knowledge helps individuals evaluate financial products and make rational financial decisions that could influence their financial resilience and well-being, while financial behavior relates to financial action determined by the individuals' financial attitude, which is a predisposition to behave in a particular manner. The previous argument is supported by Potrich, Vieira, and Mendes-Da-Silva's (2016) findings, who built and compared three models that evaluated the FL of 534 university students in southern Brazil.

FI is a multidimensional concept without a unanimously accepted definition. Early definitions of FI focused on the availability of financial services. Claessens (2006, p. 210) defined FI as “the availability of a

supply of reasonable quality financial services at reasonable costs…". More recently, Sarma and Pais (2011, p. 3) defined FI as "a process that ensures the ease of access, availability and usage of the formal financial system for all members of an economy." Pesqué-Cela, Tian, Luo, Tobin, and Kling (2021) broaden the concept of FI to include the quality of financial services dimension which reflect whether financial services meet customers' needs, the choices available to customers, customers' protection, and customers' awareness of financial services.

*2.2 Financial Literacy and Financial Inclusion: Empirical Results*

Reviewing the previous literature revealed that this research relates to two strands of studies. One of these strands deals with the studies conducted to assess FL levels in different contexts. Using data from the Standard & Poor's rating services global financial literacy survey, Klapper and Lusardi (2020) assessed FL of 150,000 adults in 140 countries. The authors calculated the FL index based on respondents' answers about four basic financial knowledge: interest rates, interest compounding, inflation, and risk diversification. They found that globally, only one out of every three adults possess financial literacy, meaning they know at least three of four financial concepts. They also found a disparity in the FL level between the economic blocs and even between the countries in the specific economic block. At the demographic level of respondents, the results showed that women, the poor, and the young are less financially literate in developed and developing countries. Karakurum-Ozdemir, Kokkizil, and Uysal (2019) studied FL in Mexico, Lebanon, Uruguay, Colombia, and Turkey. The authors used data from the financial capability survey conducted in 11 countries worldwide. FL level was assessed based on answers to five financial concepts questions correctly. The results showed that in all countries, females are less financially literate, and individuals with university degrees are more financially literate compared with individuals who do not have high school degrees. The study also reported that Uruguay has the highest FL level while Turkey has the lowest.

In the UAE, Douissa (2020) examined the association between socioeconomic and demographic factors and FL of the University of Sharjah students. The author measured FL by multidimensional score covering three dimensions: financial knowledge, behavior, and attitude. Generally, the results showed that only (31%) of students are financially literate. The results also demonstrated that male students, postgraduate students, students majoring in scientific fields, students who studied the personal finance course, students with higher CGPAs, and students with high family incomes are more financially literate. In another study, Ergün (2017) assessed the financial knowledge of a sample of 418 university students in five European countries using an online survey of 18 questions. Ten questions were used to measure correspondents' financial knowledge, such as interest rate and inflation, time value of money, risk diversification, and insurance. The results showed that male students are more knowledgeable than females. Among personal financial knowledge questions, the highest-scoring question was in insurance, followed by saving accounts, and the lowest was in the time value of money. In Turkey, Sarigül (2014) examined the relationship between the FL level and the demographic characteristics of 1,127 undergraduate students in three Turkish universities.

The authors used a questionnaire containing 22 questions about five FL concepts: general knowledge, saving and spending, banking, risk and insurance, and investing. The results showed that university students' FL level is generally low. Among FL concepts, saving and spending have the highest percentage of correct answers, while investing has the lowest percentage. The regression results showed that male students are more knowledgeable than female students and social sciences majors are more financially literate than those of educational sciences, health sciences, and natural sciences. Using an adapted version of the OECD/INFE toolkit, Oseifuah et al. (2018) measured the FL level of 342 undergraduate university students in the northern region of Ghana. The results showed that more than half of participating students are financially literate. The results also found that male students, students in the first year, and students with the highest parental income are more likely to be financially literate among participating students.

The second strand focuses on the previous studies that have empirically investigated the association of FL with an individual's financial decisions and behavior. Ćumurović and Hyll (2019) studied the association between FL and self-employment in Germany. The authors measured the FL index based on eleven questions, three covering basic financial knowledge and the rest covering advanced financial knowledge. A strong correlation was found between basic and advanced FL and the probability of self-employment. In the United States, Kim, Anderson, and Seay (2019) examined the role of financial knowledge on the financial behavior of Millennials. They found that Millennials exhibited a lower level of financial knowledge than households. They also reported a positive influence of financial knowledge on short and long-term financial behavior. Based on a sample of adult financial customers in Zimbabwe, Murendo and Mutsonziwa (2017) analyzed the impact of Fl on saving decisions. The authors measured FL by questions about respondents' financial attitudes, confidence, and knowledge. The study found that FL positively affects formal and informal saving behavior. The same result was reported by Morgan and Long (2020), who examined the relationship between FL and savings behavior in Loas.

Bucher-Koenen et al. (2021) examined the impact of FL on stock market participation and the FL gender gap in the Netherlands. They measured FL with three questions that assessed the responders' knowledge regarding compound interest, inflation, and risk diversification. The authors found a positive influence of FL on stock market participation. They also found that women have lower FL than men, negatively affecting their stock market participation. In the same context, Sivaramakrishnan, Srivastava, and Rastogi (2017) demonstrated a positive association between subjective and objective FL and the intention to invest in the stock market in India. In Tunis, Mouna and Jarboui (2015) examined the relationship between FL and the level of portfolio diversification using data from 256 individual small investors in the stock exchange. They measured FL with eleven questions covering respondents' knowledge about different financial topics. A positive impact of FL on the number of assets in the investors' portfolios was found. Unlike the previous result, Anwar, Khan, and Rehman (2017) found an insignificant effect of FL on investors' portfolio diversification in the Pakistan Stock

Exchange. Lin, Hsiao, and Yeh (2017) conducted a study in Taiwan and empirically proved that FL is positively associated with the demand for life insurance.

In continuation, and according to the previous literature review, it is concluded that there is no study investigating the association between FL and FI among university students in Jordan. Therefore, this study fills this gap in the literature. Accordingly, the following hypothesis was formulated:

***Hypothesis:*** financial literacy positively impacts financial inclusion among university students in Jordan.

## 3. Data and Methodology

*3.1 Research Sample and Instrument*

This study focuses on university students in Jordan. According to the recent statistics of the Ministry of Higher Education and Scientific Research in Jordan, the number of students in the 2022 academic year reached 379,000. Accordingly, the sample size required from the given population of 397,000 students is 665 (Krejcie & Morgan, 1970). Data was collected in March 2023 using an online questionnaire. Multilevel stratification was used in sampling as the following. First, three geographical regions in Jordan were selected: the northern, middle, and southern regions. Second, one university was selected from each region: Yarmouk University from the northern region, the University of Jordan from the middle region, and Mutah University from the southern region. The selected universities are the largest in terms of the number of students in their regions and offer academic programs in humanities, science, and health. Then, 1,000 students were randomly selected from each university. An invitation message was sent to the student's official email offered by the university, attached with a link to the questionnaire. Of the 3000 questionnaires sent to the students, 950 were returned and usable, allowing statistical inference on Jordan's university student population.

The survey of adult financial literacy developed by OECD/INFE was adopted in this study and translated into Arabic to fit the Jordanian context (OECD, 2020). The adapted questionnaire includes questions about FI and FL and questions about the demographics of students: gender, age, year of study, and major type. Four professors in the field of business tested the questionnaire's validity, and their recommendations were incorporated into the questionnaire. Moreover, the questionnaire's reliability was tested by conducting a pilot study with 60 respondents, and further adjustments were made. The result of Cronbach's alpha test (0.76) reflects an acceptable reliability and internal consistency level.

FI is measured by an overall score based on answers to seven questions indicating the respondents' holdings and active consumption of financial products, which reflects a broader concept of FI from the consumer's perspective. The resulting FI score ranges from 0 to 7. FL score is an overall score consisting of three scores: financial behavior, financial knowledge, and financial attitude. In the financial knowledge section,

there are multiple-choice and true/false questions, while the financial behavior and attitude sections utilize Likert scale questions with a five-point agreement scale.

The financial knowledge score is assessed through the completion of seven questions that assess respondents' comprehension of fundamental financial concepts, encompassing interest rate calculations, compound interest rates, risk and return evaluation, the impact of inflation, and the advantages of financial diversification. This resulting financial knowledge score can vary from 0 to 7. The financial behavior score is computed based on the answers provided to nine questions pertaining to various aspects of household budgeting, saving, prudent expenditures, bill settlements, attentiveness to financial matters, long-term financial goals, and borrowing, with a range between 0 and 9. The financial attitude score is established by evaluating the respondent's responses to three questions regarding their perspectives on money, saving, and spending, with a scale from 1 to 5. Consequently, FL's overall score is derived from the summation of the scores for financial knowledge, financial behavior, and financial attitude, which ranges from 0 to 21.

*3.2 Econometric Model*

This section attempts to determine whether the formulated hypothesis—that financial literacy positively affects financial inclusion among university students in Jordan—is confirmed by using regression and correlation analysis of the FI and Fl scores. This study's analytical framework is an econometric model, which enables a thorough analysis of the correlation between these vital factors. Table 1 reports the results of the regression analysis.

**Table 1**

Parameter estimates of the regression analysis.

**Parameter Estimates**

| | | Estimate | Std. Error | Wald | df | Sig. | 95% Confidence Interval | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Lower Bound | Upper Bound |
| Threshold | [FISCORE = 0] | -40.780 | 1.205 | 1146.174 | 1 | 0.000 | -43.141 | -38.419 |
| | [FISCORE = 1] | -36.818 | 1.106 | 1107.952 | 1 | 0.000 | -38.986 | -34.650 |
| | [FISCORE = 2] | -33.697 | 1.091 | 953.115 | 1 | 0.000 | -35.837 | -31.558 |
| | [FISCORE = 3] | -30.892 | 1.078 | 821.519 | 1 | 0.000 | -33.005 | -28.780 |
| | [FISCORE = 4] | -27.328 | 1.036 | 696.440 | 1 | 0.000 | -29.358 | -25.298 |
| | [FISCORE = 5] | -23.748 | 0.987 | 579.005 | 1 | 0.000 | -25.682 | -21.814 |
| | [FISCORE = 6] | -20.216 | 0.823 | 603.647 | 1 | 0.000 | -21.828 | -18.603 |
| Location | [FLOVERALLSCORE=0] | -42.373 | 1.287 | 1084.764 | 1 | 0.000 | -44.894 | -39.851 |
| | [FLOVERALLSCORE=1] | -42.502 | 1.353 | 986.260 | 1 | 0.000 | -45.155 | -39.850 |
| | [FLOVERALLSCORE=2] | -39.353 | 1.286 | 935.699 | 1 | 0.000 | -41.874 | -36.831 |
| | [FLOVERALLSCORE=3] | -38.837 | 1.207 | 1035.454 | 1 | 0.000 | -41.203 | -36.472 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| [FLOVERALLSCORE=4] | -37.322 | 1.166 | 1023.856 | 1 | 0.000 | -39.609 | -35.036 |
| [FLOVERALLSCORE=5] | -35.335 | 1.128 | 981.785 | 1 | 0.000 | -37.546 | -33.125 |
| [FLOVERALLSCORE=6] | -35.358 | 1.101 | 1032.103 | 1 | 0.000 | -37.515 | -33.201 |
| [FLOVERALLSCORE=7] | -35.183 | 1.113 | 1000.002 | 1 | 0.000 | -37.364 | -33.003 |
| [FLOVERALLSCORE=8] | -32.994 | 1.098 | 902.490 | 1 | 0.000 | -35.146 | -30.841 |
| [FLOVERALLSCORE=9] | -32.600 | 1.090 | 893.724 | 1 | 0.000 | -34.738 | -30.463 |
| [FLOVERALLSCORE=10] | -31.418 | 1.095 | 823.064 | 1 | 0.000 | -33.565 | -29.272 |
| [FLOVERALLSCORE=11] | -30.007 | 1.079 | 774.067 | 1 | 0.000 | -32.121 | -27.893 |
| [FLOVERALLSCORE=12] | -29.059 | 1.094 | 705.118 | 1 | 0.000 | -31.204 | -26.914 |
| [FLOVERALLSCORE=13] | -26.563 | 1.080 | 604.424 | 1 | 0.000 | -28.681 | -24.445 |
| [FLOVERALLSCORE=14] | -25.614 | 1.022 | 627.981 | 1 | 0.000 | -27.617 | -23.611 |
| [FLOVERALLSCORE=15] | -26.101 | 1.068 | 597.693 | 1 | 0.000 | -28.193 | -24.008 |
| [FLOVERALLSCORE=16] | -23.900 | 1.115 | 459.369 | 1 | 0.000 | -26.086 | -21.715 |
| [FLOVERALLSCORE=17] | -21.949 | 0.985 | 496.077 | 1 | 0.000 | -23.881 | -20.018 |
| [FLOVERALLSCORE=18] | -22.595 | 1.111 | 413.882 | 1 | 0.000 | -24.772 | -20.418 |
| [FLOVERALLSCORE=19] | -20.328 | 0.000 | | 1 | | -20.328 | -20.328 |
| [FLOVERALLSCORE=20] | 0.125 | 0.000 | | 1 | | 0.125 | 0.125 |
| [FLOVERALLSCORE=21] | 0[a] | | | 0 | | | |
| [Q4MAJORTYPE='Social Sciences'] | 0.224 | 0.195 | 1.310 | 1 | 0.252 | -0.159 | 0.606 |
| [Q4MAJORTYPE='Educational Sciences'] | 0.195 | 0.208 | 0.886 | 1 | 0.347 | -0.211 | 0.602 |
| [Q4MAJORTYPE='Health Sciences'] | 0.390 | 0.199 | 3.827 | 1 | 0.050 | -0.001 | 0.780 |
| [Q4MAJORTYPE='Natural Sciences'] | 0[a] | | | 0 | | | |
| [Q1GENDER='Female'] | -0.024 | 0.136 | 0.031 | 1 | 0.861 | -0.291 | 0.243 |
| [Q1GENDER='Male'] | 0[a] | | | 0 | | | |

Link function: Logit.

a. This parameter is set to zero because it is redundant.

Table 2 reports Spearman's rho correlational analysis for the variables in our study.

**Table 2**

The Spearman correlation matrix of the variables in the study.

**SPEARMAN'S CORRELATION MATRIX**

| | | GENDER | MAJOR TYPE | FL OVERALL SCORE | FI SCORE |
|---|---|---|---|---|---|
| **GENDER** | Correlation Coefficient | 1.000 | - | - | - |
| | Sig. (2-tailed) | - | - | - | - |

| | | | | | |
|---|---|---|---|---|---|
| **MAJOR TYPE** | Correlation Coefficient | 0.012 | 1.000 | - | - |
| | Sig. (2-tailed) | 0.711 | - | - | - |
| **FL OVERALL SCORE** | Correlation Coefficient | 0.047 | -0.059 | 1.000 | - |
| | Sig. (2-tailed) | 0.145 | 0.067 | - | - |
| **FI SCORE** | Correlation Coefficient | 0.047 | -0.054 | .894** | 1.000 |
| | Sig. (2-tailed) | 0.148 | 0.094 | 0.000 | - |

Note: ** Correlation is significant at the 0.01 level.

## 4. Results and Discussion

Building on the groundwork material offered in the previous section, the regression analysis findings in this study offer critical insights into the relationship between financial literacy (FL) and financial inclusion (FI) among university students in Jordan. The econometric model provides an in-depth look at these characteristics, which includes the financial literacy score as a significant predictor, major type, and gender as independent factors.

First and foremost, the regression analysis reported in Table 1 clearly shows that the financial literacy score is a highly significant predictor of financial inclusion scores among the sampled university students. This result supports the preceding research premise that greater financial inclusion is correlated with higher levels of financial literacy. A significant Spearman's rho correlation coefficient of about 89.4%, as reported in Table 2, which shows a strong and statistically significant positive association between financial literacy and financial inclusion ($P<0.01$), further emphasizes the strength of this relationship. The crucial role that financial literacy plays in promoting financial inclusion among university students in Jordan is highlighted by this solid empirical data.

The analysis also explores gender differences in financial literacy scores, as illustrated in Figure 1. The financial literacy study showed that men generally performed slightly better, although it is essential to highlight that this difference did not achieve statistical significance. This result suggests that, in the Jordanian environment, gender may not be a reliable factor influencing the degree of financial literacy among university students. However, further research into this conclusion is necessary to fully comprehend the complex causes underlying gender-related disparities in financial literacy.

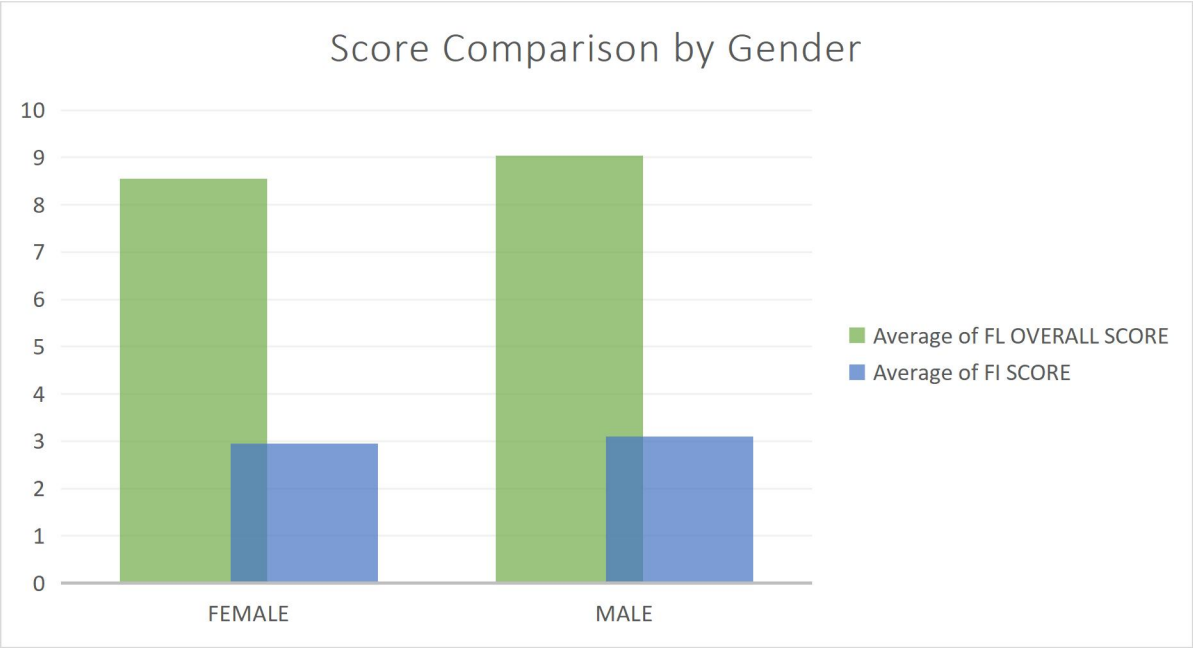


**Figure 1.**

Comparison of FL and FI scores based on gender.

Additionally, a fascinating pattern was discovered when scores on many significant forms of financial literacy were compared, as illustrated in Figure 2. Notably, compared to their peers in other areas, students majoring in social sciences were more inclined to score better on the financial literacy survey. This finding suggests that the type of academic concentration may significantly influence a person's level of financial literacy. This finding creates opportunities for further investigation into the causes of these discrepancies and the possibility of therapies catered to particular significant types.

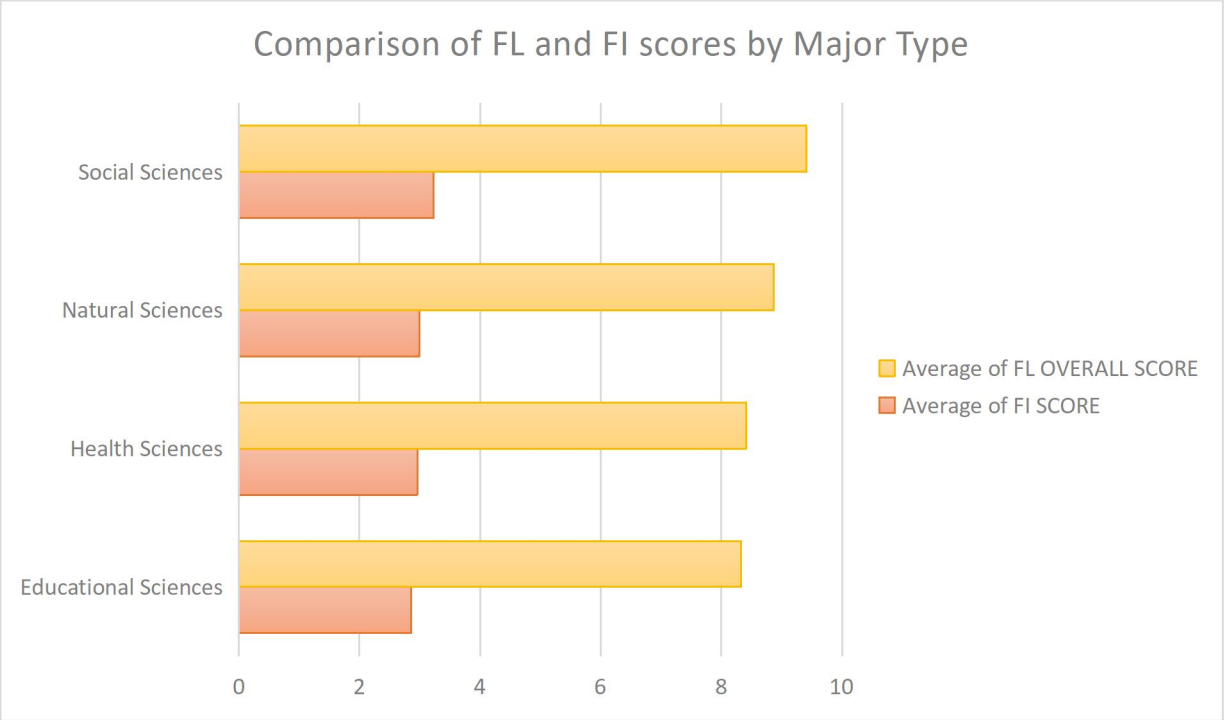


**Figure 2.**

Comparison of FL and FI scores based on significant type.

Finally, the results of the goodness-of-fit tests used to determine the statistical significance of the regression model were convincing. Compared to the null model, the McFadden value, calculated at 0.509, indicates a significant improvement in model fit. The model's ability to account for changes in financial inclusion scores depending on financial literacy, major type, and gender is highlighted by this. The statistically significant model underlines the value of the econometric technique in illuminating the nuanced relationship between financial literacy and financial inclusion, thus enhancing the study's validity.

In conclusion, the regression analysis's findings offer solid empirical support for the favorable relationship between financial literacy and financial inclusion among Jordanian university students. While identified, gender-based disparities in financial literacy did not achieve statistical significance, necessitating further research. The impact of academic majors on financial literacy further emphasizes the need for customized interventions. The statistically significant regression model strengthens the study's contributions to academic research and the development of public policy in financial inclusion and education.

## 5. Conclusion

The association between financial inclusion (FI) and financial literacy (FL) among university students in the Jordanian environment has been thoroughly investigated in this research study. This work has produced invaluable insights and additions to the body of literature on financial education and inclusion through the rigorous application of econometric methodologies, such as regression analysis and correlation analyses. The

main conclusions of this study play a crucial role in illuminating the complex interactions between FL and FI among Jordanian university students. The empirical results categorically support the critical hypothesis that financial literacy scores serve as a significant predictor of financial inclusion scores among the selected student population. This outcome supports the growing global conversation on the significance of financial education and emphasizes the crucial role that financial literacy plays in fostering financial inclusion.

It is crucial to emphasize that while gender-based differences in financial literacy scores were seen, these disparities did not achieve statistical significance in this study. This result raises the possibility that other contextual factors may significantly impact the degree of financial literacy among Jordanian university students. It is necessary to conduct more studies to understand these variables and create gender-specific financial education programs entirely. The investigation also found intriguing differences in financial literacy ratings among students with various academic disciplines. Notably, majors in the social sciences demonstrated a higher degree of financial literacy, raising concerns about the impact of academic specialization on financial knowledge and abilities. This finding emphasizes the necessity for specialized financial education programs that take into account the particular requirements of students pursuing distinct majors.

By providing empirical support for the relationship between financial literacy and financial inclusion among Jordanian university students, this study contributes considerably to the growing body of literature on financial education and inclusion. Policymakers, educators, and other stakeholders working to improve financial education programs and advance financial inclusion initiatives should consider the findings. This study highlights the significance of cultivating financial knowledge and skills among the youth population in Jordan and abroad because financial literacy continues to be a critical driver of financial well-being and economic stability.

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